

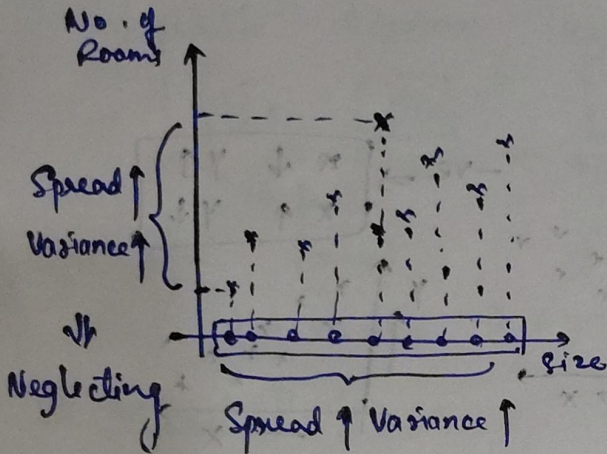
PCA Geometric Intuition:

{Dimensionality Reduction}

Housing Dataset

Size of House | No. of Rooms | Price

or



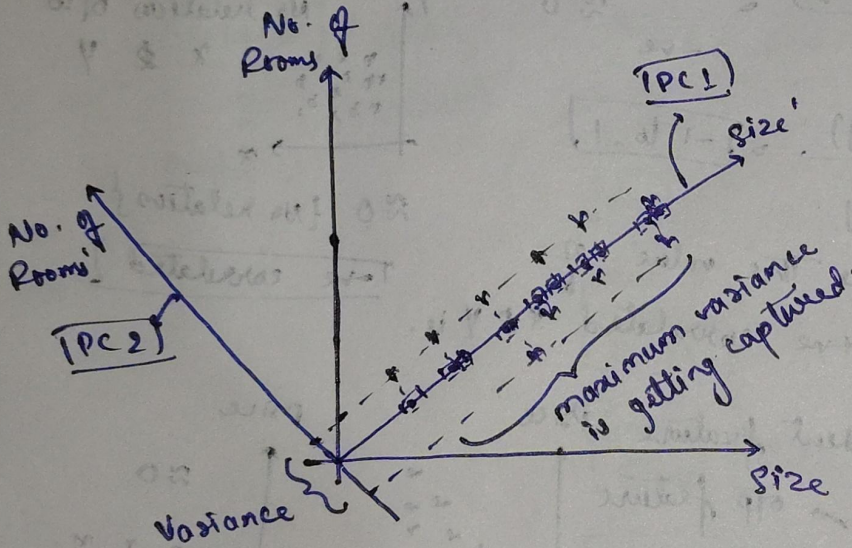
PCA

2 dimension $\rightarrow$ 1 dimension

2D $\rightarrow$ 1D

Loss of information

(No. of Rooms)
Feature Extraction



Eigen decomposition on Matrix

Transformation

2D $\rightarrow$ 1 Dimension

much information is not lost.

2 Dimension

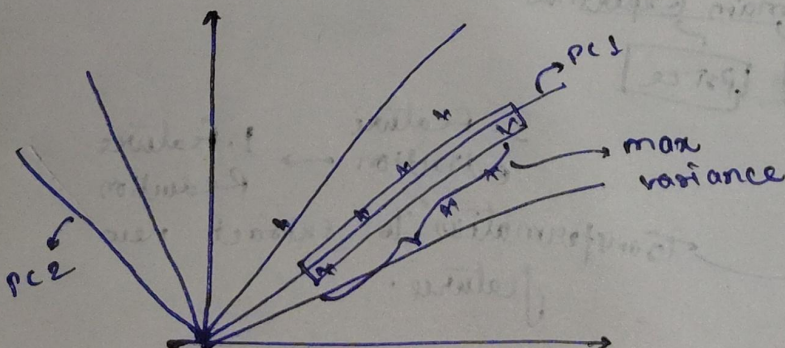
PC1, PC2

$\text{Var}(PC1) > \text{Var}(PC2)$

3 Dimension

PC1, PC2, PC3

$\text{Var}(PC1) > \text{Var}(PC2) > \text{Var}(PC3)$



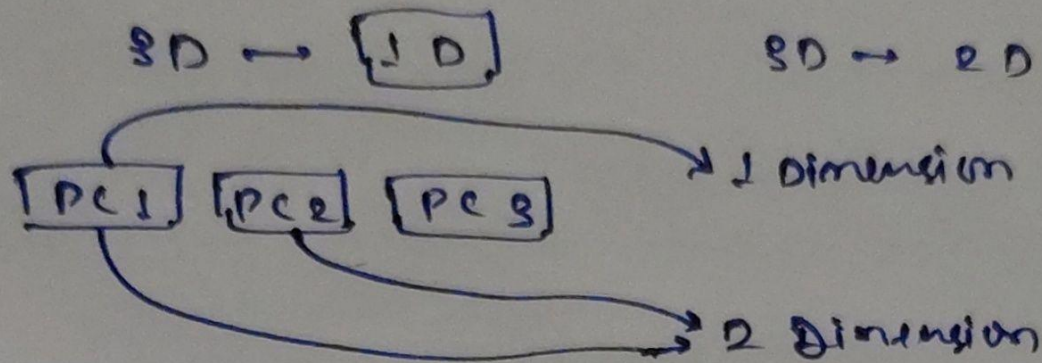
2D $\rightarrow$ 1D

Select

2 Best

principle component

To get the best principle component which captures maximum variance.



$$\text{Var}(PC_1) > \text{Var}(PC_2) > \text{Var}(PC_3)$$